Notation

Var	Meaning	Var	Meaning
Y	Total output / GDP	y = Y/L	Output per worker
K	Capital stock	k = K/L	Capital per worker
L	Labor / # workers	A or Ā	TFP or ideas stock
s	Saving rate (0–1)	d	Depreciation rate
n	Pop. growth rate	N̄	Total population
ℓ	Scientist share	z̄	Scientist productivity
g(X)	Growth rate of X	π	Inflation rate

Growth Rate Rules

$g(x/y) = g(x) - g(y)$
$g(x \cdot y) = g(x) + g(y)$
$g(x^a) = a \cdot g(x)$
$y_t = y_0(1+g)^t$ Rule of 70: doubling time ≈ 70/g

Compute average growth rate: $\bar{g} = (y_t/y_0)^{(1/t)} - 1$. e.g. g=2%/yr → doubles every 35 years.

GDP & Measurement

$Y = C + I + G + NX$

C≈70%, I≈17%, G≈17%, NX<0 for U.S. Government transfers (Social Security, welfare) NOT in G.
Investment I = new physical capital only. Buying stocks is NOT investment in GDP accounting.

$Y_N = P \cdot Y_R \rightarrow g(Y_R) = g(Y_N) - \pi$

Deflate nominal by inflation to get real. PPP adjusts for cross-country price level differences.

Cobb-Douglas Production Function

$Y = \bar{A} \cdot K^{1/3} \cdot L^{2/3} \rightarrow y = \bar{A} \cdot k^{1/3} \quad (k \equiv K/L)$
$g(y) = g(A) + ⅓ \cdot g(k) \qquad g(Y) = g(A) + ⅓g(K) + ⅔g(L)$

CRS: exponents sum to 1. DMR: each exponent < 1 (adding more K or L alone hits diminishing returns).

$MPK = ⅓ \cdot \bar{A} \cdot (L/K)^{2/3} = r$ (interest rate)
$MPL = ⅔ \cdot \bar{A} \cdot (K/L)^{1/3} = w$ (wage)
TFP residual: $\bar{A} = Y / (K^{1/3} \cdot L^{2/3})$

Ā↑ → entire production curve shifts up: y↑, w↑, r↑. Ā differs enormously across countries — main driver of income gaps.

Solow Model

$\Delta K = sY - dK$ (fixed L, no pop. growth)
$\Delta k = sy - (n+d)k$ (with pop. growth n, $k \equiv K/L$)

(n+d)k = 'total depreciation per worker': capital wears out at rate d AND new workers dilute it at rate n.

Steady state: set Δk = 0 → sĀk^⅓ = (n+d)k → solve:

$k^* = (s\bar{A} / (n+d))^{3/2}$
$y^* = \bar{A}^{3/2} \cdot (s/(n+d))^{1/2}$
$k^*/y^* = s/(n+d)$ ≈ 4 for U.S. (s=0.24, n+d=0.06)

Comparative statics — all level effects, LR growth always = 0:

Shock	k*	y*	SR g(y)	LR g(y)	Mechanism
s↑	↑	↑	+	0	saving curve shifts up
d↑	↓	↓	−	0	depreciation line steeper
n↑	↓	↓	−	0	depreciation line steeper
Ā↑	↑	↑	+	0	both curves up
K destroyed	—	—	+ (recover)	0	steady state unchanged
Immigration	—	—	− then +	0	temporary k dilution

K destroyed / immigration: steady state UNCHANGED → full LR recovery. Parameter shocks (s,d,n,Ā) → permanent new k.*

Transition dynamics:

The further k is below k*, the faster g(k) and g(y). Economy always converges to k* from any starting point.

Post-WWII Japan & Germany: capital destroyed → far below k → rapid catch-up growth. Singapore: near k*, slow TFP growth.*
KEY LIMIT: No sustained LR growth in y without ↑Ā. Higher saving rate → level boost only, not permanent growth.*

Growth Accounting

$g(Y/L) = g(A) + ⅓ \cdot g(K/L) +$ labor-hours term

TFP = g(y) − ⅓·g(k) − hours term. TFP is the residual — explains most of cross-country income differences.

Period	g(y)	g(k)	TFP	Interpretation
US 1948–73	3.3%	0.9%	2.1%	Golden age — strong TFP
US 1973–95	1.5%	0.7%	0.6%	Productivity slowdown
US 1995–02	3.0%	1.3%	1.2%	IT/internet revival

Singapore 1966–90: g(y)=4.2% but TFP≈0.2% → almost pure capital accumulation → possibly unsustainable.

Romer Model (Jones 1e–5e — THIS version is on exam)

Three core equations:

$Y_t = A_t \cdot L_{yt}$ (goods sector — no capital)

$\Delta A_t = \bar{z} \cdot A_t \cdot L_{at}$ (ideas sector)

$L_{at} + L_{yt} = \bar{N}, \quad L_{at} = \ell \cdot \bar{N}, \quad L_{yt} = (1-\ell) \cdot \bar{N}$

Solving:

$y_t = Y_t / \bar{N} = A_t \cdot (1-\ell)$ (income per person)

$g[A] = \Delta A / A = \bar{z} \cdot \ell \cdot \bar{N}$ (constant — no diminishing returns!)

$A_t = A_0(1 + g[A])^t$

$y_t = A_0 \cdot (1-\ell) \cdot (1+g[A])^t$ ← sustained exponential growth forever

On ratio scale: straight line, slope = g[A] = z̄ℓN̄. No transition dynamics. Always on balanced growth path.

Comparative statics:

Shock	Level of y	g[A]	Time path on ratio scale
N̄ ↑	unchanged	↑	slope steepens immediately, no level drop
ℓ ↑	↓ (drops)	↑	level drops first, then steeper slope
z̄ ↑	unchanged	↑	slope steepens immediately, no level drop
A_0 ↑	↑ (parallel)	—	parallel upward shift, same slope

ℓ ↑: (1−ℓ) falls immediately → y drops (fewer farmers). But z̄·ℓ·N̄ rises → faster idea growth. Short-run pain, long-run gain.

A_0 ↑ or z̄ ↑: no level drop because these don't remove workers from goods production.

Why Romer produces sustained growth (Solow cannot):

ΔA = z̄·A·L has exponent 1 on A → constant returns to ideas in idea production → g[A] never decays. Ideas are non-rival (usable everywhere simultaneously); capital is rival (can only be used in one place). Non-rivalry eliminates diminishing returns.

Solow vs. Romer

	Solow	Romer
Engine of growth	Capital K	Ideas A
Returns to engine	Diminishing (exp < 1)	Constant (exp = 1 on A)
Long-run g(y)	0 (without ↑Ā)	z̄·ℓ·N̄ > 0 forever
Steady state	k* (transition dynamics)	Balanced growth path
Rival good?	Yes → DMR	No → no DMR
Policy lever	↑s, ↓d, ↓n → level ↑	↑ℓ, ↑z̄, ↑N̄ → growth rate ↑

Key Big-Picture Lessons

1. Capital alone can't explain cross-country income gaps — the ⅓ exponent dilutes capital differences: 6% less k → only 2% less y.
2. TFP (A) explains most of the gap. E.g. k=8% of US but y=20% of US → TFP_X/TFP_US = 0.20/0.08^(⅓) ≈ 46%.
3. Saving ↑ raises the level of y* permanently but NOT the long-run growth rate. Level effect only in Solow.
4. Ideas are non-rival → no diminishing returns → sustained LR growth. Capital is rival → DMR → growth stops at k*.
5. Institutions (Acemoglu–Robinson 2024 Nobel): inclusive institutions → property rights, rule of law → high TFP.
6. Transition dynamics: countries further below k* grow faster (conditional convergence). Post-WWII Japan/Germany example.
7. Growth accounting: TFP = residual. High g(y) with low TFP = capital-driven growth, may not be sustainable (Singapore).

Solow Diagram — Drawing & Interpreting

Axes: investment/depreciation on y-axis, k (capital per worker) on x-axis.
Saving curve: sy = sĀk^⅓ — concave (bowed down) due to diminishing returns to K.
Depreciation line: (n+d)k — straight line through origin, slope = n+d.
k* = intersection. y* = point on production function Āk^⅓ above k*.
w* = gap between y* and r·k* (labor's share). r = slope of tangent to Āk^⅓ at k*.

Shock	Which curve moves?	Direction	New k*
s ↑	Saving curve	Shifts UP	→ right (↑)
d or n ↑	Depreciation line	Pivots UP steeper	→ left (↓)
Ā ↑	Both curves	Saving curve more	→ right (↑)
K destroyed	Neither	Dot moves left	Same k* (recovers)

Exam Quick Checks

✓ Growth rules: g(Y) = g(A) + ⅓g(K) + ⅔g(L). g(y) = g(A) + ⅓g(k). g(Y/L) = g(Y) − g(L).

✓ Steady state: k* = (sĀ/(n+d))^(3/2). y* = Ā^(3/2)·(s/(n+d))^(½). Compare s/(n+d) across countries.

✓ TFP cross-country: Ā_X/Ā_US = (y_X/y_US) / (k_X/k_US)^(⅓). Use this to back out TFP from data.

✓ Romer: g[A] = z̄ℓN̄. ℓ ↑ → immediate level drop AND permanent growth rate increase (two effects simultaneously).

✓ Solow LR: g(y) = 0 for all shocks. All effects are level effects only.

✓ Nominal→Real: g(Y_R) = g(Y_N) − π. e.g. nominal +7%, inflation +3% → real +4%.

✓ Capital-output ratio: k*/y* = s/(n+d). U.S.: 0.24/0.06 = 4. Cross-country comparison.

Exam rules: open book, 1 page front+back notes, calculator, ruler allowed. No internet, no devices.